Chapter 1

Introduction

Text before the first section. With some inline mathematics $f(x) = \int_0^x e^{-t^2} dt$. That looks nice.

And some displayed mathematics:

$$f(x) = \int_0^x e^{-t^2} dt.$$

Chapter 2

Sequences

Motivation and context remarks.

2.1 Initial definitions

Definition 2.1.1 Sequence. A sequence is a list of elements, indexed by the positive integers.

In these notes we will almost exclusively be considering sequences of real numbers. But mathematicians could talk of sequences of any sort of mathematical object.

We can use a single letter, x say, decorated with an index subscript, to denote the elements of a sequence, as in

$$x_1, x_2, x_3, \ldots.$$

This sequence could be denoted as a whole by the notation $\{x_n\}_{n=1}^{\infty}$ or just simply $\{x_n\}$, if the indexing is understood.

Variations might be to use the *non-negative integers* for the indexing, so the first element of the sequence would be x_0 rather than x_1. Or perhaps start the indexing at some other integer. $\Diamond$

This definition will be sufficient for our purposes, but more formal versions are possible. Some authors would define a sequence of real numbers as a function $f : \mathbb{Z}^+ \to \mathbb{R}$, i.e. f is the function from the index values to the corresponding sequence elements. So then the list/sequence of elements would be

$$f(1), f(2), f(3), \ldots.$$

An advantage of this would be to integrate our definitions with the already existing concept of a mathematical function, and not have to rely on introducing a new concept of *list*.

Example 2.1.2 Some example sequences. These examples and the other ones in this section show some basic sequences and different ways to use the notation for the sequence element and index.

1. The sequence of positive integers themselves, $1, 2, 3, 4, \ldots$. We could denote this as $\{z_m\}_{m=1}^{\infty}$, where the defining rule is simply, for $m \geq 1$

$$z_m = m.$$

2. The Fibonacci sequence

$$1, 1, 2, 3, 5, 8, \ldots$$

denoted $\{f_n\}_{m=1}^{\infty}$, where the rule is

$$f_1 = 1, \; f_2 = 1, \quad \text{and for all } m \geq 3 \quad f_m = f_{m-1} + f_{m-2},$$

i.e. each element is the sum of the previous two elements. This type of definition for a sequence, where each element is defined in terms of previous elements, is called a *recurrence relation.*

3. The sequence of odd positive integers $1, 3, 5, 7, \ldots$, denoted by $\{a_j\}_{m=1}^{\infty}$, defined by the rule

$$a_j = 2j - 1.$$

4. The sequence of prime numbers

$$2, 3, 5, 7, 11, 13, \ldots$$

denoted by $\{p_m\}_{m=1}^{\infty}$. Unfortunately there is no easy to state *formula* known for the m^{th} prime. We can specify an algorithm to find it, and various asymptotic results exist that can approximate the value of it. It can be proved that there an infinite number of primes, so the indexing declaration is valid. See more on this in your *Number Theory and Abstract Algebra* module next year.

$\square$

Definition 2.1.3 Arithmetic sequences. An arithmetic sequence is one with a constant common difference between successive elements. So we can write such a sequence as

$$a, a + d, a + 2d, a + 3d, \ldots,$$

or $\{a_n\}_{n=1}^{\infty}$, where $a_n = a + (n-1)d$, i.e. a is the initial element and d the common difference. $\Diamond$

Definition 2.1.4 Geometric sequences. An geometric sequence is one with a constant common ratio between successive elements. So we can write such a sequence as

$$a, ar, ar^2, ar^3, \ldots,$$

or $\{g_n\}_{n=1}^{\infty}$, where $g_n = ar^{n-1}$, i.e. a is the initial element and r the common ratio. $\Diamond$

Example 2.1.5 Arithmetic and geometric sequences.

1. The sequence of odd positive integers, $1, 3, 5, 7, \ldots$ is arithmetic. It is given by $\{a_n\}_{n=1}^{\infty}$ where $a_n = 1 + (n-1)2 = 2n - 1$.

2. The sequence of powers of $1/2$, which begins $1, 1/2, 1/4, 1/8, \ldots$ is geometric. It is given by

$$h_n = \left(\frac{1}{2}\right)^{n-1} = \frac{1}{2^{n-1}},$$

with initial element 1 and common ratio $1/2$.

3. A negative example. The sequence of reciprocals of the squares, $\{s_n\}_{n=1}^{\infty}$, where $s_n = 1/n^2$ is neither arithmetic nor geometric.

4. Are there any sequences that are both arithmetic *and* geometric?

5. Compound interest awarded on investments, or charged on loans, provide examples of geometric sequences. Suppose a bank overs a savings account with an annualy compounded interest rate of $r\%$ per year on an initial deposit of £P. Then the value, v_n, of the deposit at the beginning of the n^{th} year is given by
$$v_n = P\left(1 + \frac{r}{100}\right)^{n-1}.$$

□

2.2 Monotonic and bounded sequences

Before considering the *convergence* of infinite sequences, we consider the possible increasing, decreasing and bounded behaviour of sequences.

Definition 2.2.1 An increasing sequence $\{x_n\}$ is one whose elements satisfy
$$x_1 \leq x_2 \leq x_3 \leq \ldots.$$

A decreasing sequence $\{y_n\}$ is one whose elements satisfy
$$y_1 \geq y_2 \geq y_3 \geq \ldots.$$

If the inequalities are strict inequalities then we would say the sequence is strictly increasing, or strictly decreasing, as appropriate. Or the inequalities might only hold true from some element x_k onwards, in which case we would say the sequence is increasing/decreasing *from* k, as appropriate.

The term *monotonic*, refers to a sequence that is either increasing or decreasing. Similarly for *strictly monotonic*.If one has a nice formula for a sequence then the quickest way to see of the sequence is monotonic might be to plot some elements of the sequence using a computer. There are many ways to do this. Here is a quick example of how to do this using Sage.

```
S = [(n^2 + 2)/2^n for n in range(1,50)]
list_plot(S)
```

```
f(x) = (x-1)*(x+1)*(x-2)
plot(f, (x, -3, 2), color='blue', thickness=3)
```

◊